Raising OEE on a Packaging-Machine Fleet

A Lean Six Sigma DMAIC project on the PIADE dataset - project guide and report

Stack: PostgreSQL (data engine) - Excel (DMAIC analysis, 16,600+ live formulas) - Power BI (control cockpit) - Python (optional: charts and inventory simulation)

Data: 5 industrial packaging machines, 429,394 state intervals, 132 alarm codes, Jan 2020 - Dec 2021.

Honesty note. This is a retrospective study of a public dataset. Define, Measure, Analyze and the Control design are real analysis. Improve-phase gains are projections from assumed reduction rates, not measured results. Say this plainly in any interview.

0. What is in the package and how to run it

Folder / file	What it is	When you use it
sql/01_schema_and_load.sql	Loads raw_data.csv, builds f_interval (counter deltas, shifts, production day), f_downtime_event, dimensions, data-quality view	Measure - first thing you run
sql/02_dmaic_views.sql	25 views: OEE, loss tree, Pareto, MTBF/MTTR, shift tests, FMEA inputs, SMED, inventory, Kanban, X-bar/R, capability	Measure to Control
sql/03_export.sql	Exports every view to CSV for Excel / Power BI	After 02
excel/ PIADE_LSS_DMAIC_Workbook.xlsx	28 tabs in DMAIC order; all analysis as formulas; yellow cells = assumptions	Your main working file
powerbi/ PIADE_PowerBI_measures.dax	Power Query connection, relationships, DAX measures, page plan	Control - build the cockpit
python/ (optional)	Charts in this report + Monte-Carlo inventory backtest	Only if you want to regenerate figures

Run order. Install PostgreSQL 14+, create a database called piade, put raw_data.csv in the sql folder, then run:

psql -d piade -f 01_schema_and_load.sql then psql -d piade -f 02_dmaic_views.sql then psql -d piade -f 03_export.sql

Open the workbook: the Data_Daily and Data_Shift tabs already hold the SQL extracts, so every tab works immediately. To refresh with new data, paste the new CSVs into those two tabs.

Project snapshot

Metric	Baseline	Where it comes from
Fleet OEE (A x P x Q)	51.7% (64.3% x 80.8% x 99.6%)	v_oee_summary / M2_OEE
OEE by machine	s_1 65.2% | s_2 32.2% | s_3 44.2% | s_4 44.8% | s_5 55.3%	M2_OEE
Biggest loss	Idle with no reason code: 10,266 h = 27.3% of planned time	v_loss_tree / M3
Alarm downtime	3,147 h (8.4%) from 92,005 stops; 7 of 132 codes = 81.6%	v_pareto_alarm / M4
Stop profile	MTBF 15.8 run-min, MTTR 2.1 min -> minor stops	v_reliability / M5
Package quality	4,088 DPMO = 4.1 sigma (not the problem)	M6_Sigma
Pilot s_1 X-bar/R	CL 67.3%; 2021 mean 62.7% (-4.6 pts), 18 run-rule signals	v_xbar_r_chart / C1
Projected OEE	61.9% (+10.1 pts) if assumed lever rates hold	I2_OEE_Scenario

1. DEFINE - what problem, how big, what goal

What you do

1. Write the charter in D1_Charter. Its problem statement, business case and goal are formulas, so they update from the Measure tabs.
2. Fill the team row with real roles if you present this as a simulated plant project.
3. Build the SIPOC (D2) and turn each voice-of-customer complaint into a measurable CTQ with a spec (D3).

Result

Problem statement. Between Jan-2020 and Dec-2021 the 5 packaging machines ran at 51.7% OEE against a 65% target. 27% of planned time was idle with no reason code and 8.4% was lost to 92,005 alarm stops. s_1, the highest-volume machine, is the only one whose OEE fell in 2021.

Goal (SMART). Raise fleet OEE from 51.7% to at least 60% (stretch 65%) within 6 months of go-live. Hold quality at or above 99.5% and packaging-material service level at or above 98%.

Business case. Each OEE point is worth about 0.92 million good packs a year. At an assumed INR 0.60 contribution per pack, that is roughly INR 5.5 lakh a year per point.

Why s_1 is the pilot. It carries 33% of fleet volume. It owns the single largest alarm loss (A_065, 858 h). It is the only machine whose OEE dropped in 2021.

2. MEASURE - how good is the process, and can we trust the data?

What you do

1. Run sql/01. Then query v_data_quality. This is your measurement-system analysis: machine logs cannot be re-measured, so a Gage R&R is replaced by integrity checks.
2. Understand the three data traps the SQL handles. elapsed is in milliseconds. pi and po are cumulative counters read at the END of each interval (verified: the delta tracks speed x time with r = 0.95). Counter resets must be excluded (85 negative deltas, 2,397 glitches in total).
3. Define OEE the TPM way. Scheduled downtime is excluded from planned time. Idle and alarm downtime are availability losses. Performance = units in / (rated speed x run time). Quality = units out / units in.
4. Build the loss tree (M3). Both check cells must read 0; that proves the tree adds up to OEE.
5. Run the Pareto (M4), MTBF/MTTR (M5) and sigma level (M6).

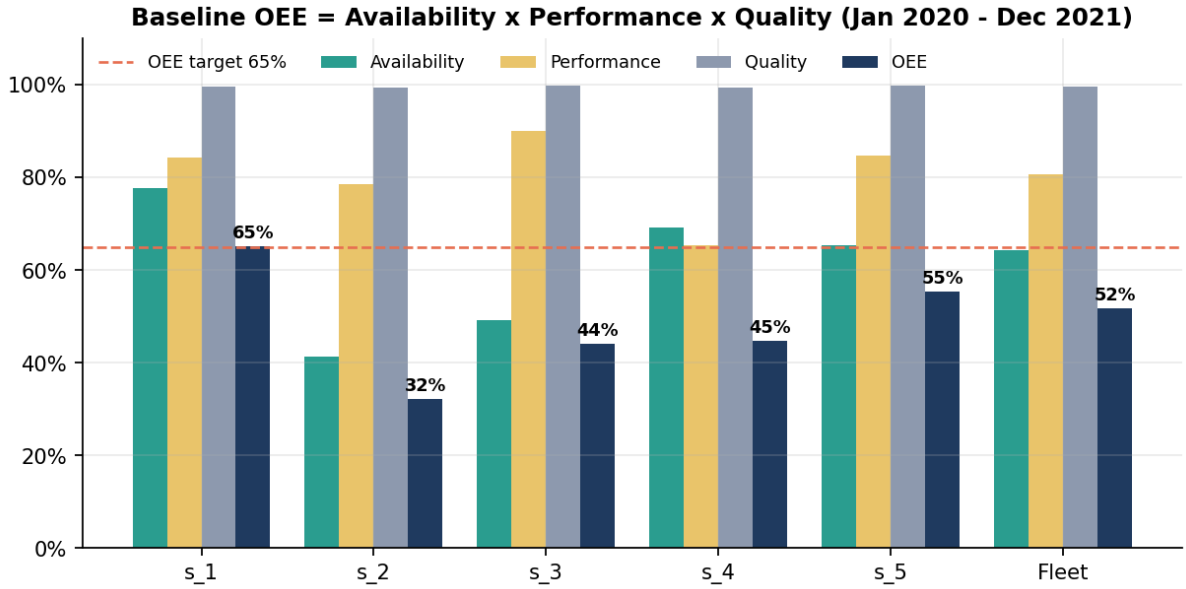


Figure 1 - Baseline OEE by machine. Only s_1 is near the 65% target.

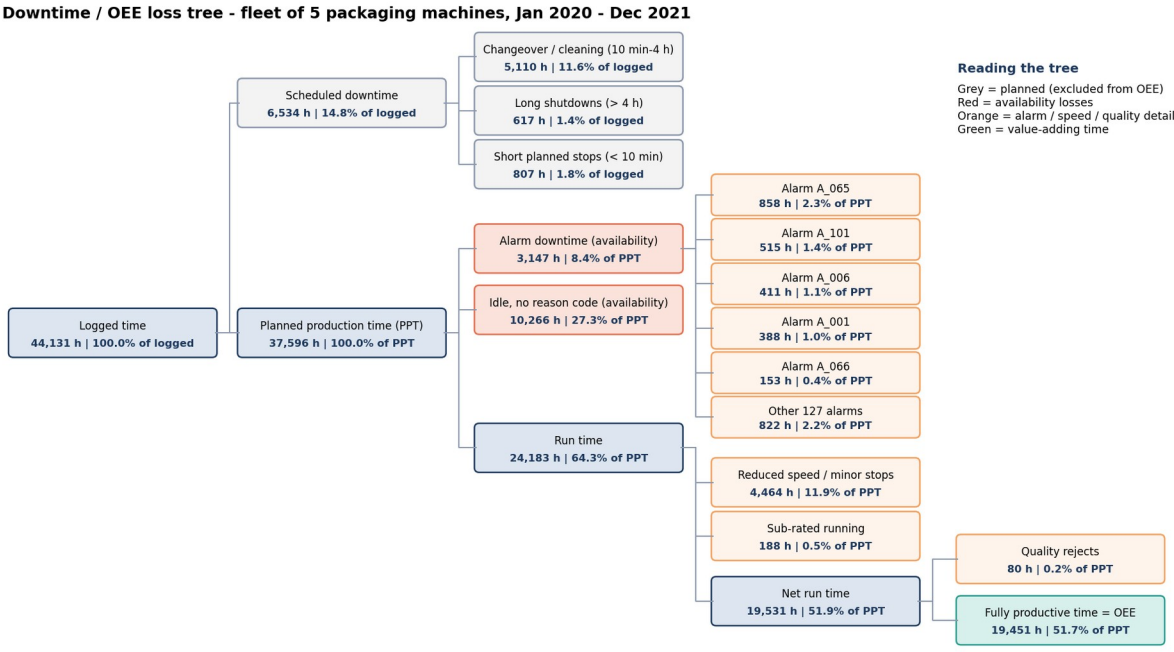


Figure 2 - Loss tree. Idle (27.3% of planned time) is 3x bigger than all alarm downtime combined.

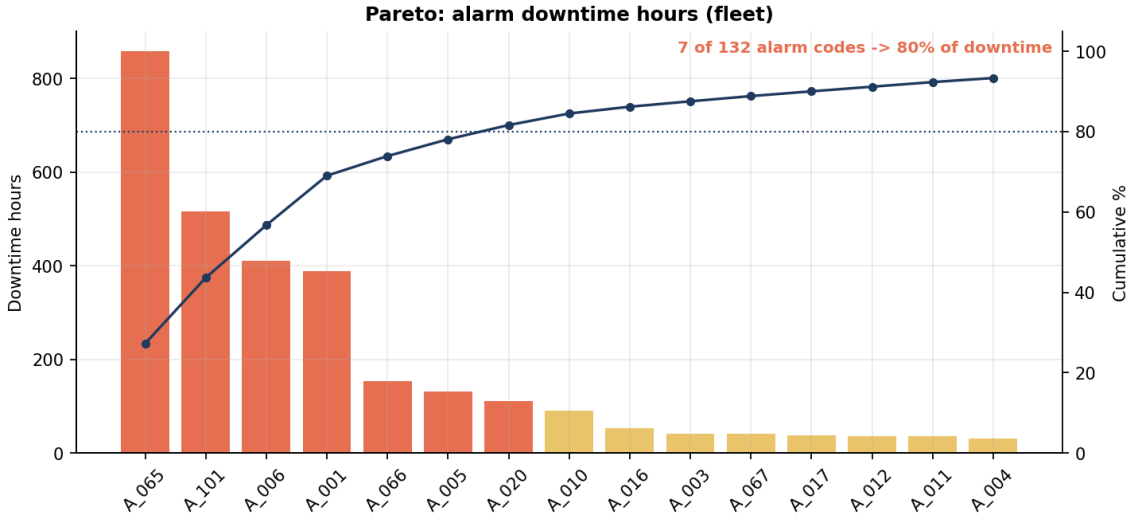


Figure 3 - Pareto: 7 of 132 alarm codes cause 81.6% of alarm downtime.

What it tells you

- **Idle is the elephant.** 10,266 h of idle carry no reason code. The data system never asks why the machine is waiting. That is a measurement failure before it is a process failure.
- **Stops are micro-stops, not breakdowns.** MTTR is 1-4 min and MTBF is 6-26 run-minutes. The right tools are Kaizen, autonomous maintenance and Poka-Yoke, not spare parts or big repairs.
- **Quality is not the problem.** Rejects run at 4,088 DPMO (4.1 sigma). Measured against the OEE target instead, s_1 is only a 1.6-sigma process. Lost time, not defects, is the real defect here.

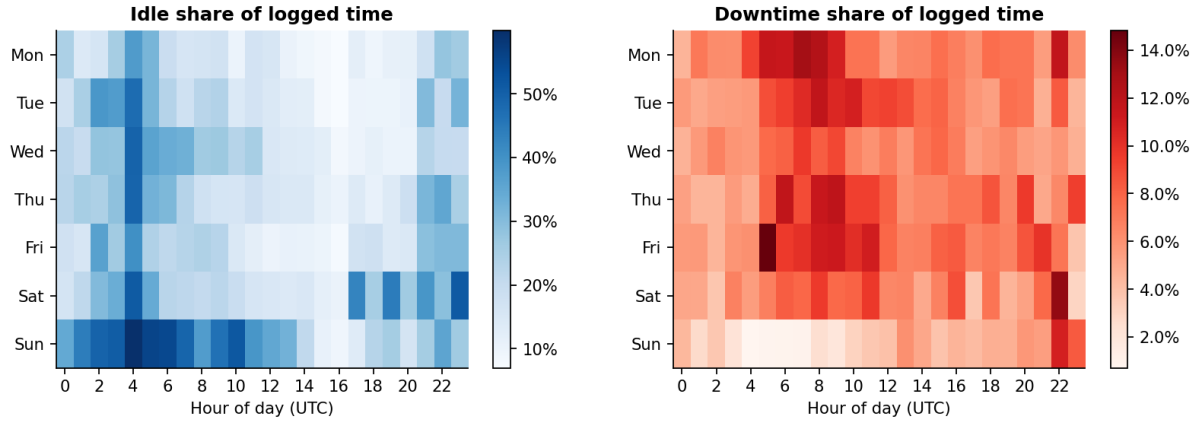


Figure 4 - Idle concentrates at night (UTC 02-05) and on Sundays; alarm downtime peaks at the start of the day shift.

3. ANALYZE - which causes actually drive the losses?

What you do

1. Brainstorm causes on a fishbone (A1). Tag every cause either [data] (verified in PIADE) or [hyp.] (must be checked on the shop floor).
2. Run 5 Whys on the two biggest problems: A_065 micro-stops and idle time.
3. Score a data-driven FMEA (A2). S comes from MTTR, O from stops per 1,000 run-hours, D from how often the stop repeats within 30 minutes.
4. Test the hypotheses with Excel-native formulas (A3): ANOVA, Welch t, chi-square and regression, all built from SUMPRODUCT, FDIST, TDIST and CHITEST so you can explain every number.

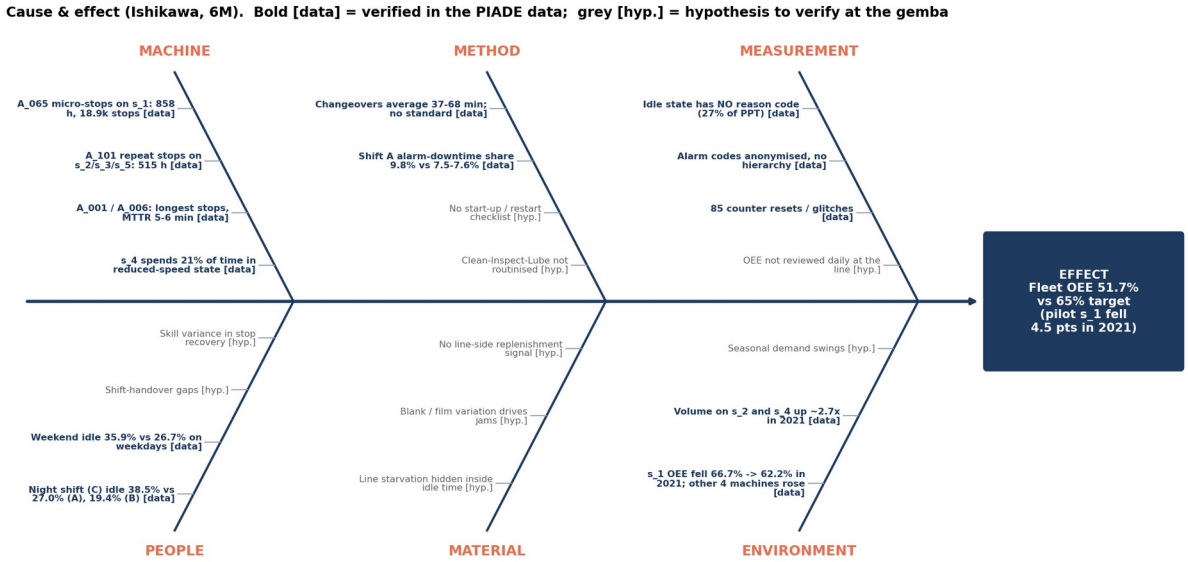


Figure 5 - Ishikawa (6M). Bold = verified in data; grey = hypothesis for the gemba.

Hypothesis test results

Test	Result	p-value	So what
H1 ANOVA: idle share by shift	A 27.0% | B 19.4% | C (night) 38.5%; F = 167.7	< 0.001	Night shift is where idle lives
H2 ANOVA: alarm-downtime share by shift	A 9.8% | B 7.6% | C 7.5%; F = 40.8	< 0.001	Start-of-day restarts cost the most
H3 Welch t: weekend vs weekday idle	35.9% vs 26.7%; t = 8.3	< 0.001	Staffing / scheduling, not the machine
H4 Chi-square: alarm mix vs shift	Chi2 = 6,386; Cramer's V = 0.19	~0	Significant but a weak effect: do not over-sell it
H5 Welch t per machine: 2020 vs 2021	s_1 -4.5 pts (drop); s_2 +9.5, s_3 +8.2, s_4 +4.0, s_5 +3.8 (rises)	all < 0.05	Pooled fleet 'decline' is a MIX effect
H6 Regression: OEE vs stops per run-h	slope -0.017 per stop/run-h; R2 = 0.14	< 0.001	Stops explain only 14% of OEE variance

The stratification lesson (H5). Pooled across machines, daily OEE looks like it fell from about 56% in 2020 to about 49% in 2021. Stratified, four of five machines actually improved. The pooled drop comes from s_2 and s_3, two low-OEE machines that only joined the data in mid and late 2020, getting a full year of weight in 2021. This is Simpson's paradox. Stratify before you conclude; it is a strong interview story.

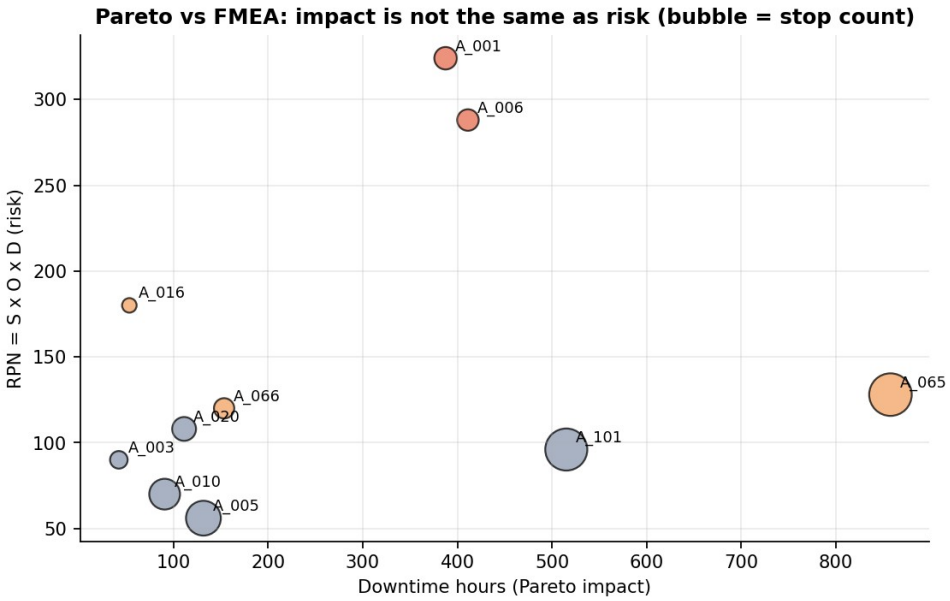


Figure 6 - Pareto ranks by impact (A_065 first). FMEA ranks by risk (A_001 RPN 324, A_006 RPN 288).

Verified root causes

- Idle time has no reason code; it concentrates at night and on weekends (H1, H3). The likely causes are staffing, material staging and scheduling. The root cause is a measurement gap.
- A_065 micro-stops on s_1 cause 858 h and 18,864 stops, and 81% repeat within 30 minutes. The trigger condition is not being removed.
- A_001 and A_006 are rarer but long, unpredictable stops, making them the highest risk. They need error-proofing and a standard repair kit.
- Speed loss is 12.4% of planned time; s_4 runs 21% of its time in reduced-speed state.
- Changeovers average 37-68 min with no standard (SMED opportunity).

4. IMPROVE - what do we change, and what is it worth?

What you do

1. Score the countermeasures in I1 (Impact x (6 - Effort)) and pick wave 1.
2. Draw the current- and future-state value stream map of the pilot s_1.
3. Quantify each lever in I2. Every assumed reduction is a yellow cell, so a reviewer can challenge any of them.
4. Design the Lean countermeasures: SMED (I3), Poka-Yoke (I4), 5S audit (I5), Kaizen event and PDCA tracker (I6), Kanban (I7).
5. Set the packaging-material inventory policy (I8): safety stock, reorder point and EOQ from the service-level target.

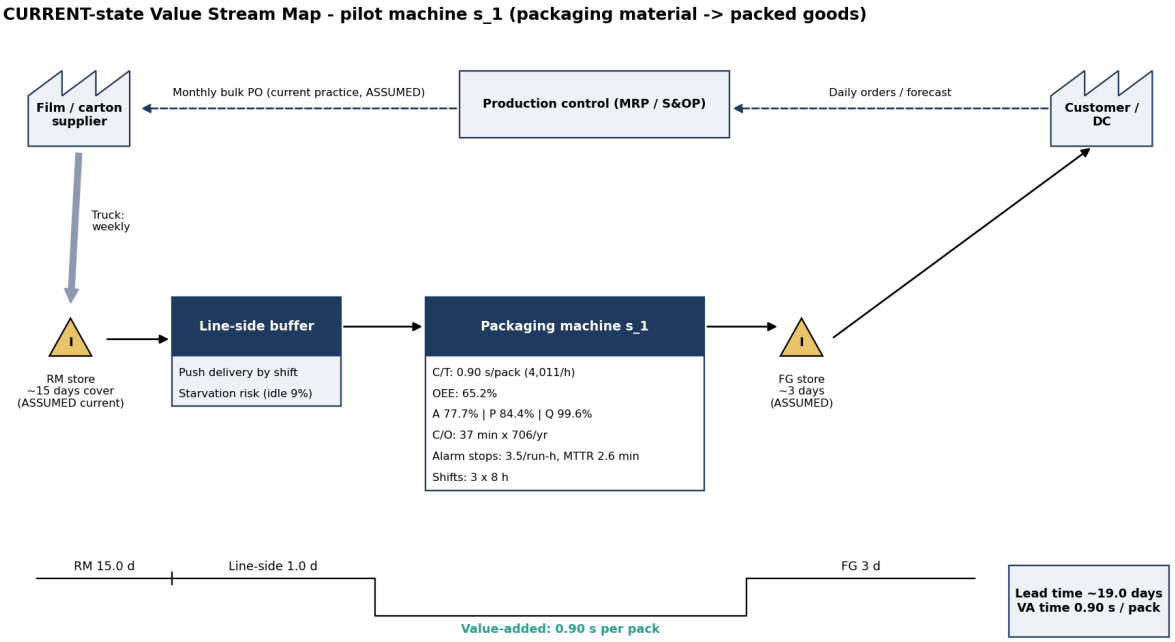


Figure 7 - Current-state VSM, s_1. Lead time about 19 days against 0.9 s of value-added time per pack. RM and FG days are ASSUMED.

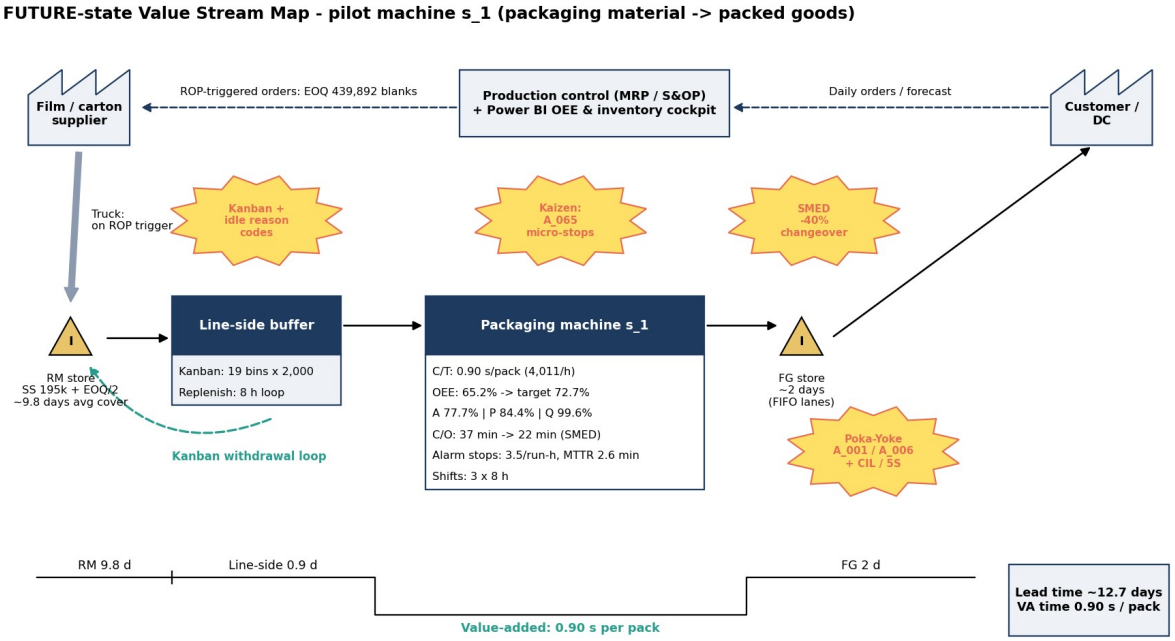


Figure 8 - Future-state VSM: ROP/EOQ supplier orders, a Kanban line-side loop, and kaizen bursts for SMED, micro-stops and Poka-Yoke.

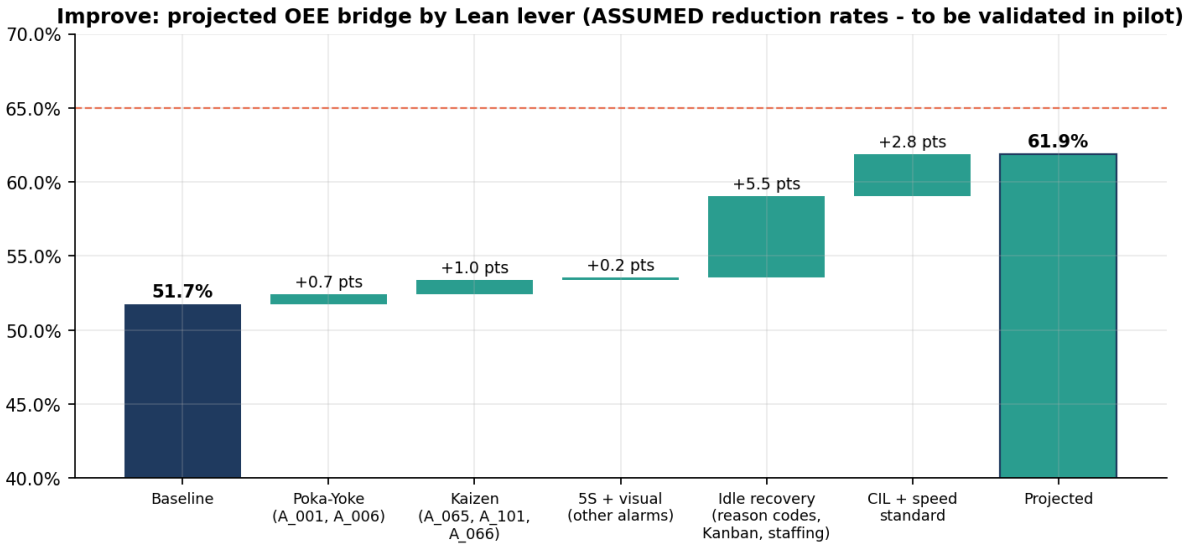


Figure 9 - Projected OEE bridge: 51.7% to 61.9%. The idle lever (+5.5 pts) beats all alarm levers combined (+1.9 pts).

Lean toolkit applied

Tool	Where it is applied	Key number
Kaizen (5-day blitz)	A_065 micro-stops on s_1; then roll out to A_101 on s_2/s_3/s_5	Target -30% stop hours; +1.0 OEE pts
Poka-Yoke	Forced idle reason code on the HMI; keyed change parts for A_001/A_006; lot scan	Idle coded >= 95%; +0.7 pts from A_001/A_006
SMED	Changeovers (planned stops 10 min - 4 h)	Fleet mean 51 min; -40% releases ~1,020 h/yr of capacity
5S + CIL	Shadow boards, Clean-Inspect-Lube, rated-speed standard	Speed loss -20% gives +2.8 pts
Kanban	Line-side packaging blanks	s_1: 19 bins x 2,000 sized on peak (rated) draw
Standard work	Night / weekend staffing and shift handover	Idle -25% gives +5.5 pts

Inventory optimisation (service-level based)

Formulas: SS = z x SQRT(L x sd_d^2 + d^2 x sd_L^2); ROP = d x L + SS; EOQ = SQRT(2DS/H). Here z = NORMSINV(service level), L = 7 days and sd_L = 1.5 days (assumed). Demand = daily packaging blanks consumed (1 blank per input package).

Machine (98% CSL)	Mean daily demand	Safety stock	Reorder point	EOQ	Days of cover
s_1	42,412	194,494	491,378	439,892	9.8
s_2	6,980	48,348	97,209	178,457	19.7
s_3	24,067	104,371	272,840	331,369	11.2
s_4	11,153	112,115	190,185	225,577	20.2
s_5	39,618	239,396	516,725	425,158	11.4

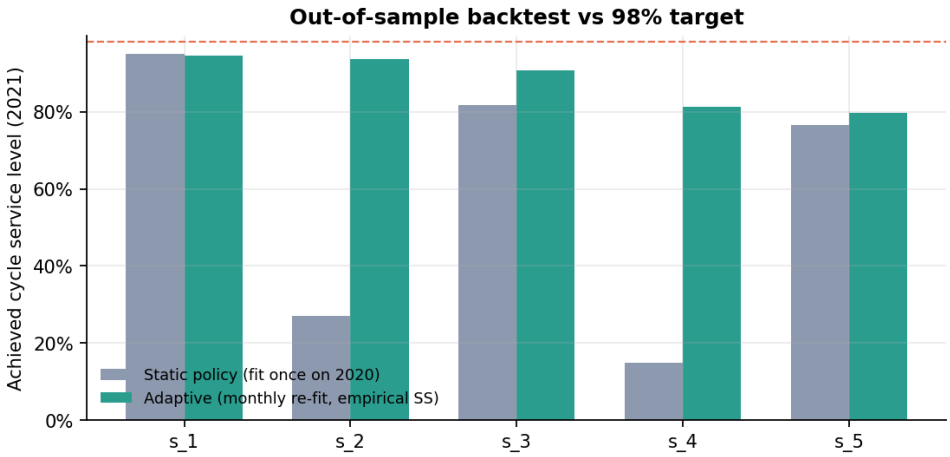


Figure 10 - Out-of-sample test: the policy was fit on 2020 and replayed on real 2021 consumption.

What the backtest proves. A policy fitted once breaks when demand shifts. s_2 and s_4 volume rose about 2.7x in 2021, and their service level collapsed to 28% and 15%. Re-fitting monthly with the larger of the normal-theory and empirical safety stock recovers them to 93.5% and 81%. Even so, lumpy daily demand keeps s_4 and s_5 below 98%. That is the honest limit of normal-theory safety stock. Spare kits for the high-risk failure modes use a Poisson reorder point: A_065 = 8, A_001 = 6, A_006 = 5 kits.

5. CONTROL - how do we hold the gain?

X-bar and R chart design

- **Characteristic.** Shift-level OEE of the pilot machine s_1.
- **Rational subgroup.** One production day, 06:00 to 06:00, with n = 3 shifts (A, B, C). The X-bar chart tracks the day-to-day level. The R chart tracks shift-to-shift inconsistency, which is exactly where Analyze found the night-shift problem.
- **Inclusion rule.** A day enters the chart only if all 3 shifts had at least 2 h of planned time.
- **Constants (n = 3).** A2 = 1.023, D3 = 0, D4 = 2.574, d2 = 1.693.
- **Phase I (2020) sets the limits.** Points beyond the first-pass limits are treated as assignable-cause candidates and removed once, then the limits are recomputed. **Phase II (2021)** is monitored against those limits.

Limit	Pass 1 (all 2020)	Revised (39 of 255 removed)
Centre line (X-double-bar)	61.5%	67.3%
UCL X	94.9%	95.9%
LCL X	28.0%	38.6%
R-bar	32.7%	28.0%
UCL R	84.2%	72.1%
Sigma within (R-bar / d2)	-	16.5%

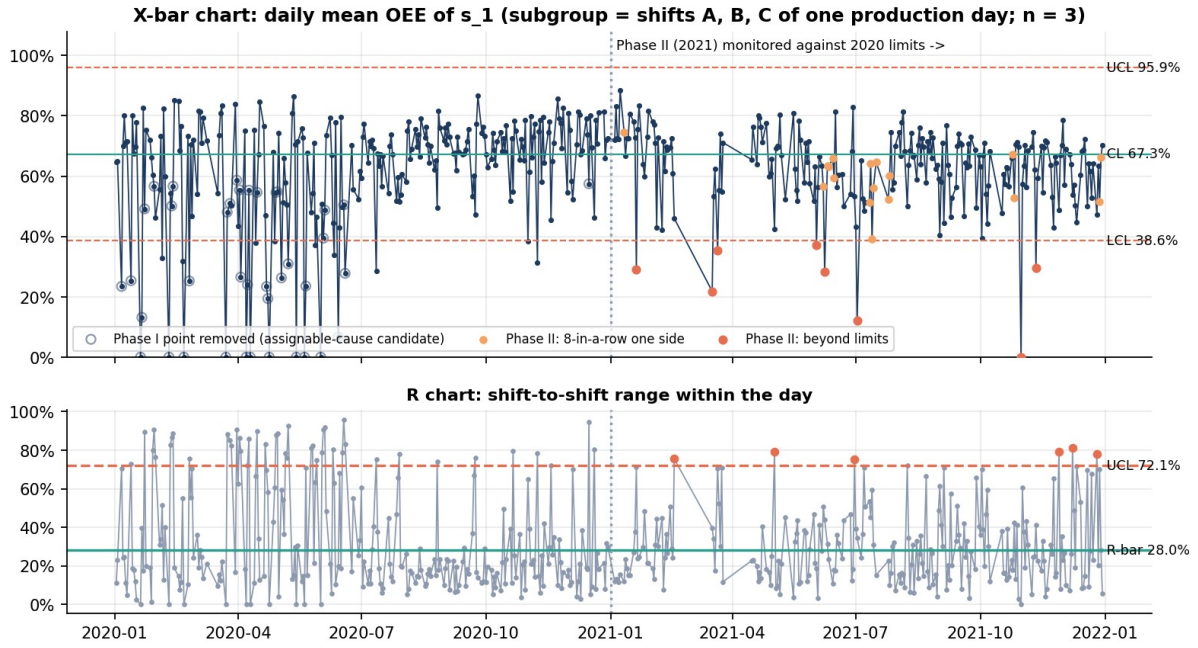


Figure 11 - X-bar and R chart for s_1. Early 2020 is chaotic; from mid-2020 the process settles. In 2021 the chart signals a downward shift.

Reading the chart. Phase II has 224 subgroups: 8 X-bar points beyond the limits, 6 R points above UCL, and 18 eight-in-a-row run-rule signals. The 2021 mean is 62.7%, 4.6 points below the revised centre line. The chart independently detects the s_1 degradation found by the t-test in Analyze. That is the point of a control chart: it would have raised the alarm in real time.

Caveat to state yourself. In a real plant you remove a Phase I point only after finding its cause. Here the 39 removed days are candidates. Most are near-zero-OEE days in the first half of 2020, which look like start-up or commissioning disruption.

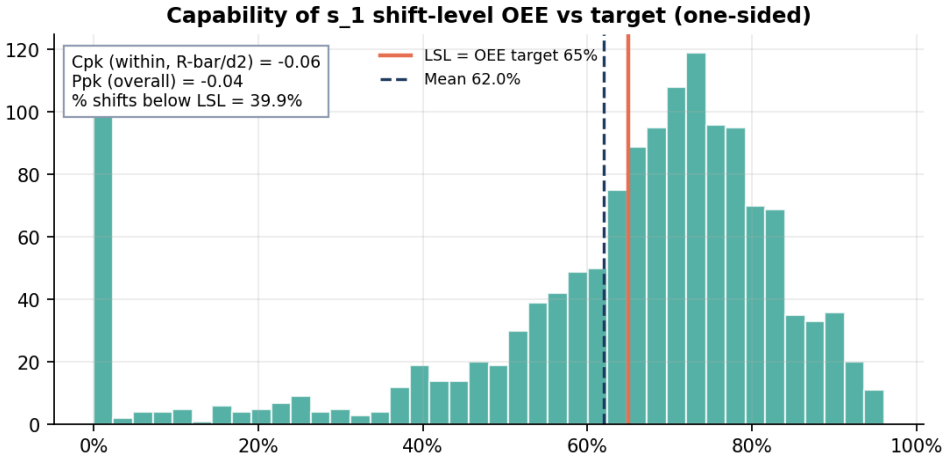


Figure 12 - Capability vs the 65% target: Cpk -0.06, Ppk -0.04, 39.9% of shifts below LSL.

Capability verdict. s_1 is not capable: it is centred below target. SPC tells you whether the process is stable; capability tells you whether stable is good enough. Here it is not, so the answer is improve, then monitor.

Control plan and reaction plan

Metric	Target	Method	Frequency	Owner
s_1 daily OEE (3-shift subgroup)	Revised limits; >= 65%	X-bar & R chart	Daily	Shift lead
Idle share by shift	<= 20%	Run chart + idle-reason Pareto	Every shift	Production mgr
% idle minutes with a reason code	>= 95%	HMI Poka-Yoke + audit	Daily	IT + shift lead
A_065 stops per run-hour	-30% vs baseline	Pareto trend	Weekly	Maintenance
Changeover minutes	-40% vs baseline	SMED standard + timing log	Each changeover	Setup lead
Material cycle service level	>= 98%	Monthly ROP/EOQ re-fit + Kanban audit	Monthly	Stores / planning
5S audit score	>= 80%	I5 checklist	Weekly then monthly	Area owner

OCAP (C4). One X-bar point below LCL: find that day's idle reasons and top alarms within the shift. R above UCL: compare the three shifts on staffing, material and changeovers. Eight in a row below the centre line: reopen the root-cause analysis.

Power BI control cockpit

Connect Power BI to PostgreSQL (the views are the model) or load the CSVs. The DAX file gives every measure: OEE, A/P/Q, DPMO and sigma, Pareto cumulative %, MTBF/MTTR, the X-bar/R lines and a service-level what-if parameter for SS, ROP and EOQ. Build six pages: OEE cockpit, loss tree (decomposition tree), Pareto & RCA, shift & idle, SPC, inventory.

6. How to execute and own this project (4-week plan)

You will be questioned on every number. Rebuild it yourself rather than presenting it as-is.

Week	Do this yourself	Proof you own it
1 - Define + SQL	Install PostgreSQL. Run 01. Read every CTE in f_interval and explain LAG, the counter delta and the production day. Rewrite the charter in your own words.	You can explain why elapsed / 1000 and why pi is read at interval end
2 - Measure	Recreate M2 and M3 formulas on a blank sheet from v_daily_machine. Reproduce 51.7% by hand. Write the Pareto query from scratch with a window function.	Loss-tree checks = 0 in your version
3 - Analyze	Redo the ANOVA and Welch t in Excel with the Data Analysis ToolPak and match A3. Find the Simpson's paradox yourself in a pivot (machine x year).	You can defend each p-value and effect size
4 - Improve + Control	Change one assumption in I2 and I8 and explain the effect. Rebuild the X-bar/R chart for s_5 by changing the machine in the SQL export. Build Power BI pages 1, 3 and 5.	A 5-minute walk-through of the X-bar/R chart

Interview talking points

- 'The biggest loss had no data behind it: 27% of planned time was idle with no reason code. My first recommendation was a measurement fix, a Poka-Yoke on data capture, before any process fix.'
- 'Pareto and FMEA disagreed. A_065 had the most hours, but A_001 and A_006 had the highest risk. I used Kaizen for the frequent stops and error-proofing for the risky ones.'
- 'The fleet looked like it got worse in 2021. Stratified, four of five machines improved. It was a mix effect, and only s_1 really declined, which is why it became the pilot.'
- 'I chose the rational subgroup deliberately: three shifts per day. The R chart then measures shift inconsistency, the exact cause Analyze found.'
- 'I backtested the inventory policy out of sample. It failed when demand shifted, so the control plan re-fits it monthly.'

Limitations (state them before you are asked)

- Improve-phase gains are projections; nothing was implemented.
- Alarm codes are anonymised, so physical root causes remain hypotheses.
- Shift times assume UTC 06/14/22 boundaries; the plant's real shift pattern is unknown.
- Costs, lead times, changeover identification and RM/FG days on the VSM are assumptions, all listed on the Inputs tab.
- OEE is bounded between 0 and 1 and skewed, so Cpk and Ppk are indicative rather than exact.